CSC 4111 Final Project

Applied Network Traffic Analysis using Machine Learning

Using machine learning techniques to
analyse network traffic patterns and detect anomalies that could indicate potential
security breaches

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Abstract

Albeit our increased technology-based society creates better avenues of communication and standards of living, there is an undisputed increase in the disruptions of security. Subsequently, increased network traffic and the development of artificial intelligence require new ways to detect intrusions, analyse malware behaviour, and categorise internet traffic and other security aspects. Machine learning (ML) shows effective capabilities in solving the upcoming network problems. ML helps to distinguish between the different traffic attributes such as packet count and size, packet inter-arrival time, packet send–receive ratio, etc. The aim of this paper is to demonstrate hands-on experience in analysing network traffic vulnerabilities, threats, and security weaknesses while discussing different ML algorithm approaches for traffic analysis. This project will assist my client Dr. Masud Rana with forming a better way of conducting network traffic analysis for his company. Random Forest (RF) is the algorithm discussed and experimented with in this paper by using the Scikit-learn environment and Wireshark to collect network traffic datasets containing packet flows with different features for training, validating, and testing the ML model. The aim was achieved by examining the features impacting Brute Force SSH and FTP attacks using the Random Forest ML technique and utilising a data set that includes updated network attacks and simulates real-world traffic flow.

Introduction

The widespread advancements of technology targeting the Internet of things (IoTs) and the Industrial Internet of Things (IIoT) is becoming more unavoidable with time. Consequently leading to an increase in cybercrime damages in households, public and private sectors creating easier access for hackers. An approach which helps combat against attacks over the network is Intrusion detection analysis (ID). Machine learning (ML), including deep learning (DL), has already created innovative traffic analysis capabilities by providing the ability to understand network traffic behaviour and patterns, and distinguish between normal and abnormal traffic[1] [2,6,7]. Network traffic analysis is an area related to machine learning techniques, which involves many efforts from computer network researchers . Analysing network traffic is one of the techniques used to detect intrusions and prevent attacks. Numerous methods exist for analysing network traffic, including self-similarity and TES, which are grounded in the study of communication systems and the identification of assaults [3]. In the meanwhile, the identification of anonymity networks forms the basis of flow analysis [4].(5). Many businesses employ machine learning to detect different forms of malicious behaviour, such as the mass registration of fictitious accounts, fraudulent transactions, and identity theft, in order to stay up with the constantly evolving attacks. Security is allegedly achieved through these applications and methods of machine learning [2]. The ability of a computer to solve problems on its own without explicit programming is known as machine learning (ML). In other words, machines may use classifiers and algorithms to handle massive datasets while continuously learning new things. The fundamental components of machine learning are classifiers, which classify observations. In

[1] Kayacik, H. G., Zincir-Heywood, A. N. and Heywood, M. I. “Selecting Features for Intrusion Detection

the meanwhile, more ML algorithms create behavioural models and then utilise those models to forecast future events based on fresh incoming data [6]. Machine learning methods are powerful because they can identify and evaluate network assaults without requiring precise definitions. In an age of vast amounts of data and a lack of skilled cybersecurity professionals, machine learning can help with typical tasks like regression, prediction, and categorization [7]. In many facets of network administration and operation, machine learning techniques have been used to maximise system performance and make better use of available resources. Additionally, patterns found in data packets are extracted via clustering and classification, which are useful for a variety of tasks like user profiling and security research [8]. The combination of machine learning algorithms and traffic analysis is a hot topic due to the power of machine learning tools that lies in detecting and analyzing network attacks without having to accurately describe them as previously defined [8].

Related Works and Background

Previous researchers also evaluated data sets using Support Vector Machine (SVM) and the Random Forest (RF) algorithm [15]. The study results indicate that a Random Forest approach takes less time to train the classifier than SVM. Their paper further proved that continued research on intrusion detection using SVM and RF is viable due to their excellent performances. Network intrusion detection systems must distinguish between hostile and benign traffic. In this paper, we apply Random Forest Regression using Python packages to examine the patterns of network attack flows from the SSH and FTP server. An RF model would be trained with features extracted from the network and transport layer headers in the traffic flows. According to [6,9] the RF model achieved a prediction rate of 97.37% in recognizing the user activities over short temporal windows. However, this system assumes that users are performing a single activity anytime, thereby obtaining one label for each temporal window. This excludes simultaneous activities performed by one user, which is an essential requirement in modern networks.

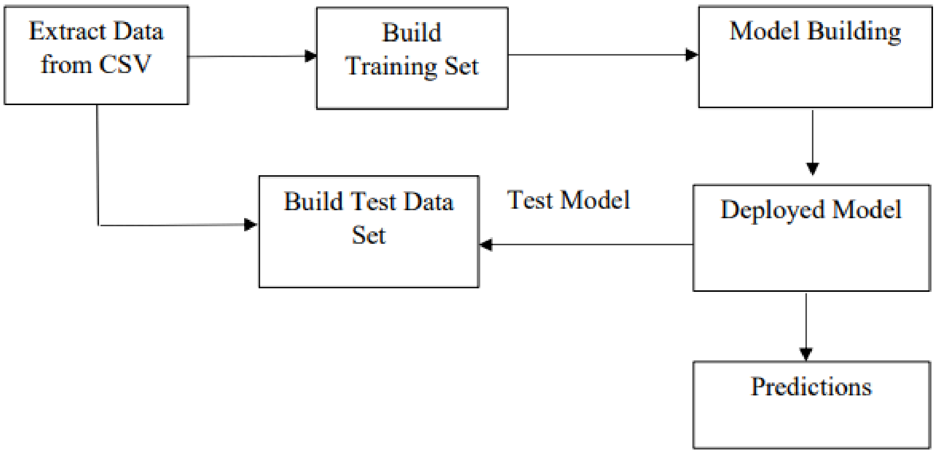
Phase 1

Methodology

This project used tools and different environments to be able to train, test and fetch results from the machine learning algorithm Random Forest to analyse network traffic.As mentioned earlier The data for this project will be injecting captured packets from wireshark from websites and youtube videos that were derived for research entailing machine learning and using wireshark. I will also be using the Scikit-learn environment for training the model machine learning algorithm.

Here are the steps taken in the project

Figure 1.1

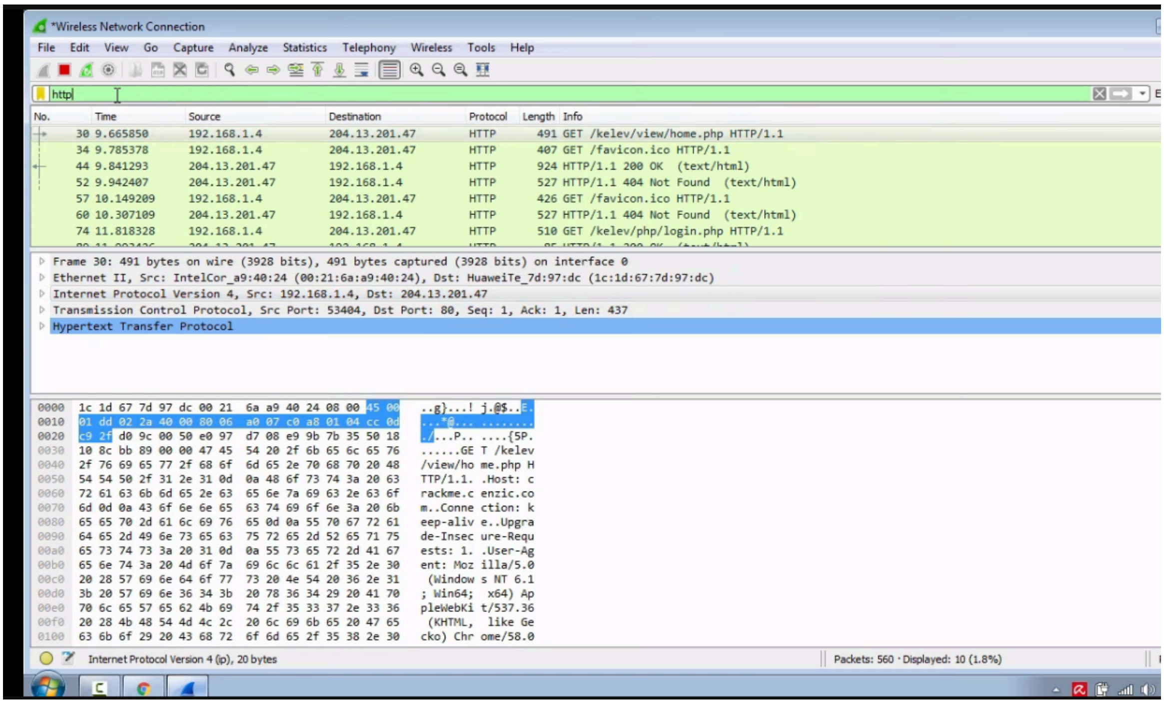


The above display of figure 1.1 shows the flow of the project and provides an overview of the steps conducted in this project.

1. Data Collection Packet Capture:

Wireshark to capture network traffic. Save the captured packets in a format that can be processed later (e.g., pcap).

Figure 1.2



In figure 1.1 we see the data packets being captured using the Wireshark environment. The highlighted blue demonstrates the focused search for traffic using the hypertext transfer protocol to narrow down the packets being searched through the network.

2. Model Training Training Split dataset into training and testing sets.

Table 1. Network Attacks and Features

Network Attack	Feature
SSH-Patator	Init Win F.Bytes
	Subflow F.Bytes
	Total Len F.Packets
	ACK Flag Count
FTP-Patator	Init Win F.Bytes
	F.PSH Flags
	SYN Flag Count
	F.Packets/s

Table 1 and 2 are taken as references by a similar study done by Altwaijry, H., and Algarny, S. "Bayesian Based Intrusion Detection System," Journal of King Saud University. The variables shown in Table 1 are said to be the best detection features for each type of attack according to [9,17]. The table explains that the features used for analyzing the patterns of SSH attacks are Init Win F.Bytes (initial window of forward bytes), Subflow F.Bytes (subflow of forward bytes), Total Len F.Packets (total length of forward packets) and ACK Flag Count. The features we used for analyzing the patterns of attacks from an FTP port are Init Win F.Bytes, F.PSH Flags, SYN Flag count and F.Packets/s (forward packets per second). A description of the features is shown in Table 2 as suggested by CICFlowMeter [19].

Table 2. Description of Features

Feature Name	Description
Init Win F.Bytes	The total number of bytes sent in initial window in the forward direction
Subflow F.Bytes	The average number of bytes in a sub flow in the forward direction
Total Len F.Packets	Total length of packets in the forward direction
ACK Flag Count	ACK (Acknowledge) Flag count
F.PSH Flags	Number of times PSH (Push) flag was set in packets travelling in the forward direction
SYN Flag Count	Number of packets with SYN (Synchronization)
F.Packets/s	Total packets in the forward direction

Phase 2 Python Pseudocode

The python scikit-learn packages were applied for machine learning prediction, as shown in Figure 2. Using Random Forest Regression, the factors impacting the SSH and the FTP attacks are independently replicated in this study Scki-learn an open-source neural network library written in Python, makes it relatively easy for developers to build and test deep learning models written in Python.

Machine Learning Model Random Forest

Random Forest is an ensemble classification and regression approach [20]. The Random Forest algorithm has been used extensively in different applications. For instance, it has been applied to prediction [21, 22], probability estimation [23], and pattern analysis in multimedia information retrieval and bioinformatics [24]. RF is a supervised learning algorithm that constructs multiple decision trees. To get an accurate and stable prediction, the final model's prediction is derived by voting on the class prediction of the individual trees in the forest. Each branch of the tree represents a possible decision, occurrence, or reaction. The RF model can be used for both classification and regression problems with a high classification rate.

$$f(x) = argmax \sum_{i=1}^{n} I\left(h_i(X)\right)$$

where $f(x)$ is the most frequently predicted class determined by voting between the outputs of $h1(x)$, …, $hn(x)$ and I is the indicator function that measures the extent to which the average number of votes for the right class exceeds the average vote for any other class. An immense margin value gives more confidence in the classification results. Using an entropy-based splitting criterion will control how each tree node splits the data. This has a significant effect on how each decision tree in the forest draws its boundaries.

```
Import packages
        from sklearn import metrics
        from sklearn.preprocessing import LabelEncoder
        from sklearn.metrics import accuracy_score
Read the security data file
Import train_test_split function
Split dataset into features and labels
        label_encoder = LabelEncoder()
        data.iloc[:,0] = label_encoder.fit_transform(data.iloc[:,0]).astype('float64')
adjust values for x and y (attributes and label)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=0)
Import Random Forest Model
Create a Gaussian Classifier
        clf=RandomForestClassifier(n_estimators=100)
Train the model using the training sets y_pred=clf.predict(X_test)
        clf.fit(X_train, y_train)
        y_pred=clf.predict(X_test)
Evaluate the algorithm
Import scikit-learn metrics module for accuracy calculation
        accuracy=accuracy_score(y_test, y_pred)
```

Phase 3

3. Data Preprocessing Feature Extraction:

[2] Shon, T., and Moon, J. “A hybrid machine learning approach to network anomaly detection.”

Extract relevant features from the captured packets. Features include source and destination IP addresses, ports, protocol types, packet size, etc. Labelling: labelled data (normal vs. anomalous), assign labels to dataset.

By using NETSCRAPPER an open source ML GUI platform to help access the model of RF and perform different tests. Below Table 3 shows the RF model and what it was about to accomplish with analysing the network.

Table 3. The precision, recall and F1-score of the RF model.

Class	Precision	Recall	F1-Score	Support
AMAZON	0.95	0.95	0.95	2958
APPLE	0.92	0.91	0.92	2977
APPLE_ICLOUD	0.97	0.99	0.98	2950
APPLE_ITUNES	0.96	0.97	0.96	3062

The precision, recall, and F1-score ratios, shown in Table 3, summarise the trade-off between the true-positive rate and the positive predictive value for our RF model using different probability thresholds. Precision represents the positive predictive value of our model, while recall is a measure of how many true positives are identified correctly, and F1-score takes into account the number of false positives and false negatives. The support metric represents the number of samples of the application class in the dataset. As shown in the table, most of the precision vs. recall values tilts towards 1.0, which means that our RF model achieves high accuracy while minimising the number of false negatives.

Phase 4 Conclusion

The location used for this experiment was between the computer labs in Hendrix and my personal computer. With my personal computer I was able to download Kali Linux which I had previously used for my Ucertify projects and was able to utilise that and download Scikit-learn. This environment helped me and allowed for me to explore the machine learning capabilities of Random Forest and how the algorithm works. I acknowledge that I made a lot of research possible to helping my findings as much of my work done on this environment could not be downloaded properly to be able to submitted. Nonetheless the experience of understanding and knowing the use of such environments alone is a good step forward towards a career in network engineering. Some of the shortcomings experienced in this project stem around machine learning being a robust topic and the length of time used to conduct some of the experiments. There would have been better research components done with greater time and resources to be able to fully grasp and understand the network analysis sequences with machine learning.

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